NLP COURSE PROJECT PROPOSAL

Design and Evaluation of an NLP-Based Task-Oriented Dialogue System

for Vietnamese Customer Support

A Comparative Study of Traditional ML, Transformer,
and LLM-based Approaches for Joint Intent Classification and Slot Filling

Course:	Natural Language Processing
Level:	4th Year
Team Size:	3 Members
Duration:	2 Weeks
Language:	Vietnamese (PhoBERT)
Dataset:	PhoATIS

1. Abstract

This project presents the design and evaluation of a Natural Language Processing (NLP) module for a Vietnamese task-oriented dialogue system targeting customer support automation. The system implements a complete pipeline: Natural Language Understanding (NLU), Dialogue State Tracking (DST), Policy, and Natural Language Generation (NLG).

The core research contribution is a **three-way comparison of NLP paradigms** for joint intent classification and slot filling: (1) Traditional ML (TF-IDF + SVM), (2) Fine-tuned Transformer (JointBERT with PhoBERT), and (3) Zero-shot LLM prompting. Experiments are conducted on the PhoATIS dataset (5,871 Vietnamese samples, 28 intents, 82 slot types). Downstream components (DST, Policy, NLG) are implemented as rule-based modules to demonstrate the full end-to-end dialogue flow.

2. Research Questions

Primary RQ: How does a fine-tuned PhoBERT model compare against traditional ML and LLM-based approaches for joint intent classification and slot filling in Vietnamese customer support dialogue?

- **RQ1:** Does PhoBERT outperform multilingual BERT on Vietnamese intent classification?
- **RQ2:** How does joint training (intent + slot simultaneously) compare to separate models?
- **RQ3:** Can a complete dialogue pipeline function effectively with high-accuracy NLU combined with rule-based downstream components?

3. Scope Definition

3.1 In Scope

#	Deliverable	Depth	Role
1	**Intent Classification — 3-way model comparison**	Deep Research	Core
2	**Slot Filling (NER) — JointBERT + PhoBERT**	Deep Research	Core
3	Dialogue State Tracking — Rule-based	Functional	Supporting
4	Policy — Rule-based decision tree	Functional	Supporting
5	NLG — Template-based response generation	Functional	Supporting
6	Demo UI — Streamlit / Gradio	Functional	Demo
7	**Evaluation report with error analysis**	Deep	Core

3.2 Out of Scope

- Real telephony / voice integration (Twilio, SIP)
- Speech recognition (ASR) and speech synthesis (TTS)
- Multi-domain support (single domain: air travel)
- Production deployment, security, microservice architecture
- Multi-language support (Vietnamese only)

3.3 Scope Justification

With a single domain and limited slot types (~8), rule-based DST achieves accuracy comparable to neural approaches (>95% on simple dialogues). This allows the team to concentrate research effort on NLU — the component with the **highest impact on overall system quality.**

4. System Architecture

The system follows the standard modular task-oriented dialogue architecture with four sequential components. NLU serves as the core research module with three competing models, while DST, Policy, and NLG are functional rule-based implementations.

USER INPUT
Raw Vietnamese text

▼

MODULE 1: NLU ★ CORE RESEARCH
Text → Intent + Slots (3 models compared)

▼

MODULE 2: DST (Rule-based)
Track conversation state across turns

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MODULE 3: POLICY (Rule-based)
Decide next system action

▼

MODULE 4: NLG (Template-based)
Action → Natural language response

▼

SYSTEM RESPONSE
Vietnamese text output to user

5. Module Specifications

5.1 NLU — Natural Language Understanding ★

Goal: Transform raw user text into structured semantic representation (intent + slots).

Input: User utterance (string), e.g., "Tôi muốn đặt vé đi Đà Nẵng ngày mai"

Output: Structured JSON: { intent, intent_confidence, slots: { slot_name: { value, position } } }

Output Example:

```
{
  "intent": "book_flight",
  "intent_confidence": 0.96,
  "slots": {
    "destination": { "value": "Đà Nẵng", "start": 24, "end": 31 },
    "time": { "value": "ngày mai", "start": 32, "end": 39 },
    "airline": null
  }
}
```

Models Compared:

Model	Type	Justification	Expected Acc
TF-IDF + SVM	Traditional ML	Baseline — fast, interpretable, no GPU	~85–90%
JointBERT + PhoBERT	Fine-tuned Transformer	Vietnamese SOTA; joint intent+slot training	~97–98%
GPT-4 / Claude (zero-shot)	LLM Prompting	Modern comparison — no training required	~70–85%

Why this comparison: SVM → Transformer demonstrates the impact of contextual embeddings. Transformer → LLM zero-shot shows that fine-tuning remains superior for domain-specific tasks. This three-way comparison is the **core academic contribution** of the project.

Evaluation Metrics:

- Intent Accuracy — percentage of correctly classified intents
- Intent F1 (macro) — averaged F1 across all intent classes
- Slot F1 (token-level) — NER performance using seqeval library
- Sentence Accuracy — percentage with both correct intent AND all correct slots
- Additional: Confusion matrix, per-intent breakdown, learning curve (20/40/60/80/100% data)

5.2 DST — Dialogue State Tracking

Goal: Maintain and update conversation state (belief state) across multiple turns.

Input: Previous belief state (dict) + current NLU output (dict)

Output: Updated belief state, e.g.: { destination: "Đà Nẵng", time: "2026-03-07", airline: "Vietnam Airlines" }

Implementation: Rule-based dictionary tracking. For each slot in NLU output: if new value exists, update state; if user requests change, clear conflicting slot.

Justification: With a single domain and ~8 slot types, rule-based DST is sufficient. Neural DST (T5, TRADE) provides marginal benefit only in multi-domain settings (5+ domains).

5.3 Policy — Action Decision

Goal: Decide the system's next action based on current dialogue state.

Input: Belief state + current intent + dialogue history

Output: Dialogue action, e.g.: { action: "request_slot", slot: "time" }

Action Space (6 actions):

- request_slot(slot_name) — ask for missing information
- confirm_booking() — confirm with user before execution
- provide_info(info) — answer user's query
- execute_action() — perform the requested action
- clarify() — ask user to rephrase when intent is unclear
- escalate_to_human() — transfer to human agent

Implementation: Rule-based decision tree. Justification: Transparent, debuggable, sufficient for single-domain. RL-based policy requires a simulator and thousands of training episodes.

5.4 NLG — Natural Language Generation

Goal: Convert structured action into natural Vietnamese response.

Input: Action dict from Policy module

Output: Vietnamese text string

Implementation: Template-based with slot filling. ~20–30 templates covering all action types. Justification: Deterministic, high quality output, no GPU cost.

6. End-to-End Dialogue Example

The following example demonstrates the complete pipeline flow across three conversation turns:

TURN 1	
User	"Tôi muốn đặt vé máy bay đi Đà Nẵng"
NLU Output	Intent: book_flight (0.96) | Slots: { destination: "Đà Nẵng" }
DST State	{ destination: "Đà Nẵng", time: null, airline: null }
Policy	request_slot("time") — time is missing
NLG Response	"Bạn muốn bay vào ngày nào?"

TURN 2	
User	"Ngày mai, Vietnam Airlines nhé"
NLU Output	Intent: inform (0.91) | Slots: { time: "ngày mai", airline: "Vietnam Airlines" }
DST State	{ destination: "Đà Nẵng", time: "2026-03-07", airline: "Vietnam Airlines" }
Policy	confirm_booking() — all required slots filled
NLG Response	"Xác nhận đặt vé Vietnam Airlines đi Đà Nẵng ngày 07/03. Đúng không ạ?"

TURN 3	
User	"Ừ đúng rồi"
NLU Output	Intent: confirm (0.98) | Slots: {}
Policy	execute_action()
NLG Response	"Đã đặt vé thành công! Mã đặt chỗ: VN20260307-001"

7. Dataset

Primary dataset: PhoATIS — the first public Vietnamese dataset for joint intent detection and slot filling, localized from the English ATIS dataset by VinAI Research. Vietnamese airports, airlines, and city names replace the English equivalents.

Dataset	Samples	Intents	Slots	Language	Domain
PhoATIS	5,871	28	82	Vietnamese	Air travel
ATIS (reference)	5,871	21	~120	English	Air travel

Data split: 70% training / 15% validation / 15% test. Preprocessing requires VnCoreNLP word segmentation (mandatory for PhoBERT tokenization).

8. Evaluation Framework

8.1 Primary Evaluation (NLU)

Metric	Description	Target
Intent Accuracy	Percentage of correctly classified intents	>95%
Intent F1 (macro)	Averaged F1 across all 28 intent classes	>93%
Slot F1 (token)	NER performance via seqeval	>94%
Sentence Accuracy	Both intent AND all slots correct	>85%

8.2 Analysis Plan (Academic Core)

The following analyses differentiate this project from a simple implementation exercise:

- **Confusion Matrix Analysis:** Identify which intents are most frequently confused and provide linguistic explanations
- **Error Analysis:** Manually examine 20+ misclassified examples, categorize error types (ambiguous intent, rare slots, OOV words)
- **Learning Curve:** Train with 20%, 40%, 60%, 80%, 100% data — plot accuracy vs. dataset size for each model
- **Slot-level Analysis:** Identify which entity types are hardest to extract and analyze reasons

8.3 Expected Comparison Results

Model	Intent Acc	Slot F1	Train Time	Inference
TF-IDF + SVM	~85–90%	N/A	<1 min	<1ms
JointBERT + PhoBERT	~97–98%	~96%	~2–3 hours	~20ms
GPT-4 zero-shot	~70–80%	~60–70%	0 (no train)	~1–2s

8.4 Secondary Evaluation (Pipeline)

- DST: Slot tracking accuracy on 50 manually created test dialogues
- End-to-end: Task completion rate (% of dialogues successfully completed)

9. Team Division & Timeline

9.1 Team Roles

Member	Role	Responsibilities
A	**NLU Research Lead**	TF-IDF+SVM baseline, PhoBERT fine-tuning, all NLU evaluations, error analysis
B	**Pipeline Engineer**	DST + Policy + NLG implementation, integration testing, demo UI (Streamlit/Gradio)
C	**Data & Report Lead**	Dataset preprocessing, LLM zero-shot evaluation, literature review, report writing

9.2 Two-Week Timeline

Day	Member A	Member B	Member C
WEEK 1			
1–2	Setup env, preprocess PhoATIS, train SVM baseline	Design DST logic, policy rules, NLG templates	Literature review, prepare dataset splits
3–4	Fine-tune JointBERT with PhoBERT	Implement DST + Policy + NLG modules	Implement LLM zero-shot evaluation
5	NLU evaluation runs	Integration: connect all 4 modules	Compile baseline results
WEEK 2			
6–7	Error analysis, confusion matrix, ablation study	E2E testing, bug fixes, build demo UI	Write methodology + results sections
8–9	Write NLU evaluation section	Demo polishing, test dialogues	Write intro, related work, conclusion
10	Review full report	Final demo check	Prepare presentation slides

10. Key References

[1] Chen et al. (2019), "BERT for Joint Intent Classification and Slot Filling" — arXiv:1902.10909. Foundational JointBERT architecture.

[2] Nguyen & Nguyen (2020), "PhoBERT: Pre-trained Language Models for Vietnamese" — EMNLP Findings. Vietnamese language model SOTA.

[3] Dao, Truong & Nguyen (2021), "JointIDSF: Intent Detection and Slot Filling for Vietnamese" — INTERSPEECH. PhoATIS dataset + PhoBERT-based joint model.

[4] Qin et al. (2019), "Stack-Propagation Framework with Token-Level Intent Detection" — EMNLP. Improved joint NLU.

[5] Weld et al. (2022), "A Survey of Joint Intent Detection and Slot Filling" — ACM Computing Surveys. Comprehensive survey.

[6] Wei & Zou (2019), "EDA: Easy Data Augmentation" — EMNLP. Data augmentation for small datasets.

[7] Bucher & Martini (2024), "Fine-tuned Small LLMs Outperform Zero-shot Generative AI Models" — arXiv:2406.08660.

[8] FitzGerald et al. (2023), "MASSIVE: 1M-Example Multilingual NLU Dataset" — ACL. 51-language benchmark including Vietnamese.

11. Tools & Technologies

Component	Tool	Justification
Language	Python 3.10+	Standard for NLP research
DL Framework	PyTorch	HuggingFace compatibility
Transformers	HuggingFace Transformers	Pre-trained models + fine-tuning
Vietnamese Tokenizer	VnCoreNLP (RDRSegmenter)	Required for PhoBERT
NER Evaluation	seqeval	Standard token-level metrics
Traditional ML	scikit-learn	SVM, TF-IDF, metrics
Demo UI	Streamlit / Gradio	Fast prototyping
Coding Assistant	Claude Code	Accelerate development

12. Expected Deliverables

1. Trained NLU models (SVM, JointBERT/PhoBERT) with saved checkpoints
2. Preprocessed PhoATIS dataset with train/val/test splits
3. Complete evaluation results: accuracy tables, confusion matrices, learning curves
4. Working end-to-end demo via Streamlit/Gradio
5. Technical report (8–12 pages): literature review, methodology, experiments, results, analysis
6. Presentation slides

13. Limitations & Future Work

13.1 Known Limitations

- Single domain only (air travel) — generalization untested
- Rule-based DST/Policy — cannot handle complex multi-turn corrections
- Limited dataset (~5,800 samples) — may not reflect real-world distribution
- Template NLG — responses lack natural diversity
- No real user testing — evaluation on synthetic dialogues only

13.2 Future Work

- Extend to multi-domain support (hotel, restaurant, e-commerce)
- Replace rule-based DST with neural approaches (TRADE, SimpleTOD)

- Add sentiment-aware escalation (detect frustrated users)
- Integrate Whisper ASR for voice-based interaction
- Deploy as web service with production API architecture